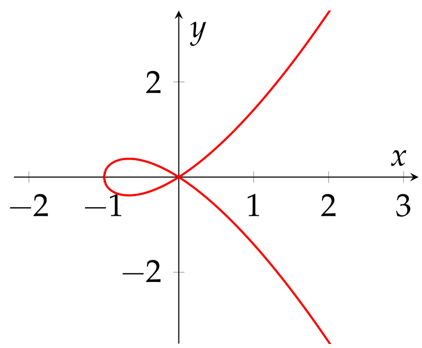
REES ALGEBRAS AND BLOWUP ALGEBRAS

PRISHITA DHARAMPAL

1. What is a Blowup?



We care about studying varieties and singularities are an obstruction to that. Intuitively, a singularity is a point in the variety where the tangent is not well-defined. For instance, in $\mathbb{A}^2$ the curve $y^2 = x^3 + x^2$ has node at the origin where two branches cross and we have two perfectly valid tangent directions, $y = x$ and $y = -x$. **TODO:** add caption to the figure

Blowups are a systematic tool for resolving singularities like this one. The idea is fairly straightforward, instead of picking one tangent direction at the singular point, we replace the point with a copy of $\mathbb{P}^1$, which parametrizes all lines through it simultaneously. The two branches then lift to the blowup as separate smooth curves, intersecting $\mathbb{P}^1$ at different points.

In general, the blowup of a variety $X \subset \mathbb{A}^n$ at a point replaces that point with a copy of $\mathbb{P}^{n-1}$, thus recording all the directions one can approach it from. So, it lives in $\mathbb{A}^n \times \mathbb{P}^{n-1}$.

Projective space $\mathbb{P}^n$ parametrizes lines through the origin in $\mathbb{A}^{n+1}$, i.e., a point $[p_0 : p_1 : \cdots : p_{n-1}]$ represents a line through 0 and $(p_0, \ldots, p_{n-1})$ in $\mathbb{A}^{n+1}$ with two points being equivalent if they differ by a non-zero scalar. Thus, a polynomial f only vanishes consistently at a projective point if $f(\lambda x) = 0$ whenever $f(x) = 0$, forcing f to be homogeneous. Therefore subvarieties of $\mathbb{P}n - 1$ are cut out by homogeneous ideals. For instance, given the (homogeneous coordinate) ring $k[x_0, \ldots, x_n]$, in its associated projective space $\mathbb{P}^n$ we have

- (0) — corresponds to all of $\mathbb{P}^n$ **TODO:** not accurate
- (x_i) — a hyperplane $\{x_i = 0\} \cong \mathbb{P}^{n-1}$
- (x_i, x_j) — a codimension-2 linear subspace $\{x_i, x_j = 0\} \cong \mathbb{P}^{n-2}$
- (f) for homogeneous irreducible f of degree d - a hypersurface $\cong \mathbb{P}^{n-d}$.

2. Graded Rings and Modules

Since subvarieties of $\mathbb{P}^{n-1}$ are cut out by homogeneous ideals we need a framework to work with them. The polynomial ring $R = k[x_1, \ldots, x_n]$ decomposes naturally by degree,

$$R = \bigoplus_{d \geq 0} k[x_1, \ldots, x_n]_d$$

where $R_d = k[x_1, \ldots, x_n]_d$ denotes homogeneous polynomials of degree d. Note that multiplication respects this decomposition,

$$R_d \cdot R_e \subseteq R_{d+e}$$

This ring $R = \bigoplus_{d \geq 0} R_d$ is called a graded ring. The polynomial ring has a natural grading by degree, but a general ring R doesn't. To extract a graded ring from R, we pick ideals $I_i \subset R$ such that the descending multiplicative chain holds,

$$R \supset I_1 \supset I_2 \supset I_3 \supset \cdots$$

where $I_i I_j \subset I_{i+j}$. When the I_i are powers of a single ideal, this chain is called a I-adic filtration, and it allows us to measure "how deeply" an element lies in I. For our purposes, I is always the maximal ideal of a local Noetherian ring R.

From this filtration we can recover the associated graded ring $\mathrm{gr}_I R$,

$$\mathrm{gr}_I R := R/I \oplus I/I^2 \oplus \cdots$$

Where the multiplication in $\mathrm{gr}_I R$ is well-defined. For $a \in I^n/I^{n+1}$ and $b \in I^m/I^{m+1}$, we pick represtatives $a' \in I^n$ and $b' \in I^m$ and define the product $ab \in I^{n+m}/I^{n+m+1}$ to be the image of $a'b'$. One can check that this is well-defined modulo I^{m+n+1}. We will now develop some theory around filtrations and graded rings that will be helpful later.

The idea of I-adic filtrations can be generalized to the case of modules.

Definition 2.1. *Given an R-module M, and an ideal $I \subseteq R$ we get the I-adic filtration of the module,*

$$M \supset IM \supset I^2 M \supset \cdots$$

and the associated graded module,

$$gr_I M := M/IM \oplus IM/I^2M \oplus I^2M/I^3M \oplus \cdots$$

with multiplication defined as follows: for $a \in I^n/I^{n+1}$ and $b \in I^m M/I^{m+1}M$, we pick represtatives $a' \in I^n$ and $b' \in I^m M$ and define the product $ab \in I^{n+m}M/I^{n+m+1}M$ to be the image of $a'b'$.

Remark 2.2. *Note that intersecting the terms of the I-adic filtration of M with some submodule $M' \subset M$ sadly generally does not give us the I-adic filtration of M'.*

The Artin-Rees lemma characterizes exactly the cases where this condition does hold. We will state it here and prove it after having developed the requisite theory.

Lemma 2.3 (Artin-Rees)**.** *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R-modules. If*

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots$$

is a I-stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \cdots$$

is also I-stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

Definition 2.4. *Let R be a ring, $I \subseteq R$ an ideal, and M an R-module. A filtration $M = M_0 \supset M_1 \supset \cdots$ is called an I-filtration if $IM_n \subseteq M_{n+1}$ for all $n \geq 0$, and I-stable if there exists m such that $M_{m+i} = I^i M_m$ for all $i \geq 0$.*

Notation. We write M_k to denote $I^k M$, and $\mathrm{gr}_I R = \bigoplus_{k \geq 0} I^k / I^{k+1}$ and $\mathrm{gr}_I M = \bigoplus_{k \geq 0} M_k / M_{k+1}$ for the associated graded ring and module respectively.

All ring maps $R \to \mathrm{gr}_I R$ factor over the quotient R/I, since any such map must send I to positive degree elements and hence $I \subset \ker(\phi_0)$, where ϕ_0 denotes the degree 0 component,

$$\begin{tikzcd} R \arrow[r, "\phi"] \arrow[d, two heads] & \mathrm{gr}_I R \\ R/I \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

However there is no interesting natural homomorphism from M to $\mathrm{gr}_I M$, but there is an interesting natural set map called the initial form.

Definition 2.5. *Let $f \in M$ and m be the greatest number such that $f \in M_m$. Then, the intial form of f, $in(f)$, is defined as,*

$$in(f) = f \pmod{M_{m+1}} \in M_m / M_{m+1} \subset grM$$

or if $f \in \cap^{\infty} M_m$

$$in(f) = 0.$$

Definition 2.6. *For a submodule $M' \subset M$ we define $in(M')$ to be the $gr_I R$-submodule of $gr_I M$ generated by $in(f)$ for all $f \in M'$.*

Remark 2.7. *Unfortunately, $in(M')$ is generally not obtained as the submodule generated by the initial forms of the generators of M'.*

For instance, take $R = k[x,y]$, $I = (x,y)$, and $J = (x^2 + y^3, xy) \subset R$ with the I-adic filtration. The initial forms of the generators are $in(x^2 + y^3) = x^2$ and $in(xy) = xy$, so the submodule generated by their initial forms is $(x^2, xy) \subset gr_I R \cong k[x,y]$. However, consider:

$$y \cdot (x^2 + y^3) - x \cdot (xy) = x^2 y + y^4 - x^2 y = y^4 \in J$$

so $in(y^4) = y^4 \in in(J)$. But $y^4 \notin (x^2, xy)$ in $k[x,y]$, since every element of (x^2, xy) has x as a factor while y^4 does not. Therefore $in(J) \supsetneq (in(x^2 + y^3), in(xy))$.

Proposition 2.8. *Let R be a ring, $I \subset R$ be an ideal, and let M be a finitely generated R-module with I-filtration $\mathcal{J} : M = M_0 \supset M_1 \supset \cdots$ by finitely generated modules M_i. The filtration $\mathcal{J}$ is I-stable if and only if the $B_I R$-module $B_{\mathcal{J}} M$ is finitely generated.*

Proof.

($\Longrightarrow$)

Assume $\mathcal{J}$ is I-stable. Then we have $M_{n+i} = I^i M_n$ for some n and all $i \geq 0$. That is, for all $k > n$, M_k is generated from M_n by multiplying powers of I. Since each $M_0, \ldots, M_n$ is finitely-generated, the union of their generating sets gives a finite generating set for $B_{\mathcal{J}}M$.

($\Longleftarrow$)

Assume $B_{\mathcal{J}}M$ is finitely generated. Since $B_{\mathcal{J}}M = \bigoplus_{n\geq 0} M_n$ is a direct sum, every element has only finitely many nonzero homogeneous components. Therefore each generator g_i is contained in $M_0 \oplus \cdots \oplus M_{k_i}$ for some finite k_i, and taking $n = \max(k_1, \ldots, k_r)$, all generators are contained in $M_0 \oplus \cdots \oplus M_n$.

Each generator g_i decomposes as a finite sum of homogeneous pieces, and the $B_I R$-submodule generated by all these homogeneous pieces contains the $B_I R$-submodule generated by $\{g_i\}$, since each g_i is a sum of its homogeneous pieces. Since $\{g_i\}$ generates $B_{\mathcal{J}}M$, the homogeneous pieces also generate $B_{\mathcal{J}}M$. Hence we may assume $B_{\mathcal{J}}M$ is generated by homogeneous elements in $M_0, \ldots, M_n$.

Since all generators live in degrees $\leq n$, any element of M_{n+i} is produced by multiplying a generator $m \in M_k$ with $k \leq n$ by some $a \in I^{n+i-k}$. Since $\mathcal{J}$ is an I-filtration, $I^{n-k}m \in M_n$, so $a \cdot m = I^i \cdot (I^{n-k}m) \in I^i M_n$. Therefore $M_{n+i} \subseteq I^i M_n$. The reverse inclusion $I^i M_n \subseteq M_{n+i}$ follows immediately from $\mathcal{J}$ being an I-filtration. Hence $M_{n+i} = I^i M_n$ for all $i \geq 0$, hence $\mathcal{J}$ is stable. **TODO:** proofread $\square$

Lemma 2.9 (Artin-Rees)**.** *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R-modules. If*

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots$$

is a I-stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \cdots$$

is also I-stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

Proof. Let

$$\mathcal{J}' : M' = M'_0 \supset M'_1 := M' \cap M_1 \supset M'_2 := M' \cap M_2 \supset \cdots$$

be the induced filtration on M'. The module $B_{\mathcal{J}'}M' = \bigoplus_{n\geq 0}(M' \cap M_n)$ is naturally a graded $B_I R$-submodule of $B_{\mathcal{J}}M$.

Since $\mathcal{J}$ is I-stable, $B_{\mathcal{J}}M$ is finitely generated as a $B_I R$-module by Proposition 2.9. Since R is Noetherian and I is finitely generated, $B_I R = \bigoplus_{n\geq 0} I^n t^n$ is a finitely generated R-algebra and hence Noetherian by the Hilbert basis theorem. Therefore the submodule $B_{\mathcal{J}'}M'$ of the finitely generated $B_I R$-module $B_{\mathcal{J}}M$ is itself finitely generated. Applying Proposition 2.9 again, $\mathcal{J}'$ is an I-stable filtration of M'. **TODO:** Proofread $\square$

Remark 2.10. *One of the important applications of associated graded constructions is when I is maximal ideal then $gr_I R$ is a graded algebra over the field R/I and if I is finitely-generated, $gr_I R$ is a finitely-generated algebra. Thus, gr_I allows us to turn arbitrary Noetherian rings into finitely-generated graded rings which are easier to study.* ***TODO:*** *Why? tbh we loved Noetherian rings all term*

3. Blow Up Algebras

Definition 3.1. *Let R be a ring, and $I \subset R$ be an ideal. The* ***blowup algebra*** *of I in R is the R-algebra,*

$$B_I R := \bigoplus_{n \geq 0} I^n t^n = R \oplus It \oplus I^2 t^2 \oplus \cdots \cong R[tI] \subset R[t]$$

where t is a formal variable tracking the degree of the ideal. The associated graded ring is obtained by quotinenting by $IB_I R$:

$$B_I R / IB_I R = gr_I R = R/I \oplus I/I^2 \oplus \cdots$$

The blowup algebra is the algebraic structure behind the geometric operation of blowing up a subvariety. Suppose R is the coordinate k-algebra of an affine variety X, and $I \subset R$ is the ideal of a subvariety $Y \subset X$. Let $a_1, \ldots, a_r$ be k-algebra generators for R and let $g_0, \ldots, g_s$ be generators of I as an ideal of R.

Proposition 3.2. *The blowup algebra $B_I R$ is a homomorphic image of $k[x_1, \ldots, x_r, y_0, \ldots, y_s]$.*

Proof. Define a ring homomorphism

$$\phi : k[x_1, \ldots, x_r, y_0, \ldots, y_s] \to B_I R, \quad x_i \mapsto a_i, \quad y_j \mapsto g_j t$$

This is surjective since any element of $B_I R$ is a finite sum of terms of the form $f \cdot gt^n$ with $f \in R$ and $g \in I^n$, and since f is a polynomial in the a_i and g is a polynomial in the g_j, every such term is in the image of ϕ. By the first isomorphism theorem:

$$B_I R \cong k[x_1, \ldots, x_r, y_0, \ldots, y_s] / \ker(\phi)$$

Since $B_I R$ is a graded ring and ϕ respects the grading by sending the x_i to degree 0 elements and the y_j to degree 1 elements, the kernel is a homogeneous ideal. $\square$

$k[x_1, \ldots, x_r]$ is the coordinate ring of $\mathbb{A}^r$, and $k[y_0, \ldots, y_s]$ is the homogeneous coordinate ring of $\mathbb{P}^s$. So $k[x_1, \ldots, x_r, y_0, \ldots, y_s]$ is the coordinate ring of the product $A^r \times A^{s+1}$, and quotienting by a homogeneous ideal in the y_j variables makes the vanishing scale-invariant in the y_j and hence cuts out a subvariety that is projective in the y-directions,

$$Z = \mathbb{V}(\ker \phi) \subset \mathbb{A}^r \times \mathbb{P}^s$$

The projection $\pi : \mathbb{A}^r \times \mathbb{P}^s \to \mathbb{A}^r$ restricts to a map $Z \to X$ which is an isomorphism away from the preimage of Y. The set Z is called the **blowup of Y in X**, and the preimage $\pi^{-1}(Y) \subset Z$ is called the **exceptional set**.

TODO: Algebraically, the exceptional set corresponds to:

$$B_I R / IB_I R = \mathrm{gr}_I R$$

The exceptional set is the projective variety associated to $\mathrm{gr}_I R$. Recall that $\mathrm{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is a graded ring, and its homogeneous prime ideals not containing all of $(\mathrm{gr}_I R)_{>0}$ correspond to points of a projective variety, exactly as homogeneous prime ideals of $k[x_0, \ldots, x_s]$ correspond to points of $\mathbb{P}^s$. This variety parametrizes all limiting directions of approach to Y inside X — the degree 1 piece I/I^2 is the space of directions, and taking its projectivization replaces Y with the set of all lines through it, one for each limiting direction of a secant line to X through Y.

We are now ready to revisit the example we looked at at the start of this note more formally,

Example: Blowing up the node of $y^2 = x^3 - x^2$. The curve $y^2 = x^3 - x^2$ has a node at the origin. The Jacobian $(-3x^2 + 2x, 2y)$ vanishes at $(0,0)$, confirming it is singular.

We blow up $Y = \{0\}$ inside $X = \mathbb{V}(y^2 - x^3 + x^2)$. The coordinate ring is $R = k[x,y]/\langle y^2 - x^3 + x^2\rangle$ and the maximal ideal of the origin is $I = \langle x, y\rangle$ (by Nullstellensatz).

The generators of R as a k-algebra are $\{x, y\}$ and the generators of I are also $\{x, y\}$. Relabeling the generators of R as x_1, x_2 and the generators of I as y_0, y_1, the blowup algebra is a homomorphic image of $k[x_1, x_2, y_0, y_1]$ by the surjection,

$$\phi : x_1 \mapsto x, \quad x_2 \mapsto y, \quad y_0 \mapsto xt, \quad y_1 \mapsto yt$$

The kernel is generated by the curve relation $x_2^2 - x_1^3 + x_1^2$ and the relation $x_1 y_1 - x_2 y_0$, which lies in $\ker(\phi)$ since $\phi(x_1 y_1 - x_2 y_0) = x \cdot yt - y \cdot xt = 0$. Since this relation is homogeneous of degree 1 in y_0, y_1, the blowup cuts out an algebraic subset $Z \subset \mathbb{A}^2 \times \mathbb{P}^1$.

4. Tangent Cone

Theorem 4.1 (Krull's Intersection Theorem)**.** *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R-module. There exists an element $r \in I$ such that*

$$(1-r)\bigcap_{k=1}^{\infty} I^k M = 0$$

Proof. Let $N = \bigcap_{k=1}^{\infty} I^k M$. We want to find $r \in I$ such that $(1-r)N = 0$. Consider the I-adic filtration on M,

$$M \supset IM \supset I^2 M \supset \cdots$$

and the induced filtration on N,

$$N \supset N \cap IM \supset N \cap I^2 M \supset \cdots$$

By Artin-Rees applied to $N \subset M$, there exists m such that for all $n \geq m$,

$$N \cap I^n M = I^{n-m}(N \cap I^m M)$$

In particular,

$$N \cap I^{m+1} M = I(N \cap I^m M)$$

But since N is the intersection of all $I^k M$ we have $N \subseteq I^{m+1}M$, so $N \cap I^{m+1}M = N$. Similarly $N \subseteq I^m M$, so $N \cap I^m M = N$. Therefore:

$$N = IN$$

Since R is Noetherian and M is finitely generated, N is a finitely generated R-module. Applying Nakayama's lemma to $N = IN$: there exists $r \in I$ such that $(1-r)N = 0$. $\square$

Corollary 4.1.1. *If R is a domain or $(R, \mathfrak{m})$ is a local Noetherian ring, and $I \subsetneq R$ then*

$$\bigcap_{k=1}^{\infty} I^k = 0$$

Proof. Let $M = R$. Since I is a proper ideal, there exists some $r \in I$ such that $r \neq 1$, so $1 - r \neq 0$. If R is a domain, we are done. If R is a local ring, $r \in I \subseteq \mathfrak{m}$ so $1 - r \notin \mathfrak{m}$, hence $1 - r$ is a unit and must be non-zero. Hence, $\cap_{k=1}^{\infty} I^k = 0$. $\square$

Corollary 4.1.2. *If R is a Noetherian local ring and $I \subsetneq R$ is an ideal then if $gr_I R$ is a domain, so is R.*

Proof. If $fg = 0 \in R$ then $\text{in}(f)\text{in}(g) = 0 \in \text{gr}_I R$. Since $\text{gr}_I R$ is a domain, $\text{in}(f)$ or $\text{in}(g)$ is zero. Without loss of generality, assume $\text{in}(f) = 0$. Then by definition, $f \in \cap_{k=1}^{\infty} I^k$ and by KIT we know that $\cap_{k=1}^{\infty} I^k = 0$. Hence $f = 0$. Thus, R is a domain. $\square$

Definition 4.2. *A line through 0 in $\mathbb{A}^r$ is determined by its direction (which corresponds to a point in $\mathbb{P}^{r-1}$). More precisely, the line through 0 and $p = (p_1, \ldots, p_r)$ corresponds to $[p_1 : \cdots : p_r] \in \mathbb{P}^{r-1}$ So a sequence of secant lines ℓ_{p_n} at a point $X \subset A^r$ is really a sequence of points $[p_n] \in \mathbf{P}^{r-1}$. A* ***limiting position*** *is a limit point of this sequence:*

$$\ell = \lim_{n \to \infty} [p_n] \in \mathbb{P}^{r-1}$$

This is well-defined because P^{r-1} is compact (over $\mathbb{C}$), so every sequence has a convergent subsequence.

The correspondence between the exceptional set in the blow up and the associated graded ring gives us a beautiful geometric consequence. Let $R = k[x_1, \ldots, x_r]/J$ and $I = (x_1, \ldots, x_r)$ where k is algebraically closed and $J \subset I$, also let X be the algebraic subset of $\mathbb{A}^r$ cut out by J,

$$X = \mathbb{V}(J) \subset \mathbb{A}^r \text{ so } 0 \in X$$

Then we define the tangent cone of X at 0 as the cone composed of all lines that are the limiting positions of secant lines to X passing through the point 0. Alternatively, the tangent cone at 0 is then the closure of the set of directions of points on X, together with the actual lines they represent in $\mathbb{A}^r$.

$$TC_0(X) = \overline{\{[p] \in \mathbf{P}^{r-1} : p \in X \setminus \{0\}\}} \subset \mathbf{P}^{r-1}$$

Since the closure captures limiting directions that arise as limits of secant lines but may not themselves be a point of $X \setminus \{0\}$.

TODO: Add figure and theorem

Theorem 4.3. *Let $R = k[x_1, \ldots, x_r]/J$ with $I = (x_1, \ldots, x_r)$ and k algebraically closed. The tangent cone $TC_0(X)$ to $X = \mathbb{V}(J) \subset \mathbb{A}^r$ at the origin is the algebraic subset:*

$$TC_0(X) = \mathbb{V}(in_I(J)) \subset \mathbb{A}^r$$

where $in_I(J) = \langle in(f) : f \in J \rangle \subset k[x_1, \ldots, x_r]$ is the ideal generated by the initial forms of all elements of J. Moreover, the coordinate ring of the tangent cone is:

$$k[x_1, \ldots, x_r]/in_I(J) \cong gr_I R$$

Proof. We sketch the key steps; for a complete proof see Harris **[?]**.

First, $\text{in}_I(J)$ is a homogeneous ideal since each $\text{in}(f)$ is homogeneous by definition, so $\mathbb{V}(\text{in}_I(J))$ is indeed a cone in $\mathbb{A}^r$.

For the geometric identification: a direction $[v] \in \mathbb{P}^{r-1}$ lies in $TC_0(X)$ if and only if v is a limit of directions $[p_n]$ with $p_n \in X \setminus \{0\}$ and $p_n \to 0$. Writing $p_n = t_n v_n$ with $t_n \to 0$ and $v_n \to v$, and expanding $f(t_n v_n) = 0$ for $f \in J$, the leading order term gives $\text{in}(f)(v) = 0$. Since this holds for all $f \in J$, we get $v \in \mathbb{V}(\text{in}_I(J))$. The converse holds by a similar limiting argument.

For the ring isomorphism, note that the natural map:

$$k[x_1, \ldots, x_r] \twoheadrightarrow \text{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$$

sending x_i to its class in I/I^2 is surjective with kernel $\text{in}_I(J)$, giving $\text{gr}_I R \cong k[x_1, \ldots, x_r]/\text{in}_I(J)$. □

Example 4.1. *For the nodal curve $y^2 = x^3 - x^2$, the lowest degree part of $f = y^2 - x^3 + x^2$ is $in_I(f) = y^2 - x^2 = (y-x)(y+x)$. So the tangent cone is $\mathbb{V}(y^2 - x^2) \subset \mathbb{A}^2$, the union of the two lines $y = x$ and $y = -x$, confirming the two tangent directions at the node. The coordinate ring is $k[x, y]/(y^2 - x^2) \cong gr_I R$.*

One can show that the ideal $\mathrm{in}_I(J) \subset k[x_1, \ldots, x_r]$ defines the tangent cone so that the coordinate ring of the tangent cone is $\mathrm{gr}_{(x_1,\ldots,x_r)}R$. **TODO:** exercise 5.3 eisenbud

5. Conclusion

6. Worksheet

Problem 1.

(1) Motivation
(2) KIT
(3) Blow Ups, Blow Up Algebras, Exceptional Sets
(4) Limiting Points and Tangent Cones
(5) Flatness
(6) Projective and Free Resolutions
(7) 'Criteria for Flatness'
(8) Rees Algebras

(1) **TODO:** Examples
(2) What makes a good family? Flatness!
    - Algebraically, this means that when the family is represented by a R-algebra S, then S is flat as a R-module.
    - Geometrically this means that when the family is represented by a morphism $\phi : X \to B$, then for each point $x \in X$ there is an affine neighborhood U of x and a affine neighborhood of $\phi(x)$ such that ϕ restricts to a map $U \to V$ and the corresponding map of coordinate rings $K[V] \to K[U]$ makes $K[U]$ into a flat $K[V]$-module.
(3) Free resolution helps "standardize" the sequence so tor can give us a measure of the faliure of exactness **TODO:** examples and definitions
(4) **TODO:** Criteria for Flatness.
(5) **TODO:** Rees Algebras as a flat family.